

Multiple Choice (10 marks)

1. Besides temperature, what other physical conditions must be taken into account for the growth of bacteria?
 - (a) Levels of oxygen, acidity or alkalinity of the medium
 - (b) Temperature changes, light exposure
 - (c) Atmospheric pressure, noise levels
 - (d) Nitrogen levels, radiation exposure
 - (e) Soil texture, light wavelength
2. Why do cells contain lysosomes?
 - (a) Lysosomes act as storage units for excess nutrients
 - (b) Lysosomes are used for energy production
 - (c) Lysosomes function in intra cellular digestion
 - (d) Lysosomes synthesize proteins for the cell
 - (e) Lysosomes regulate cell division
3. How are the variable portions of heavy chains combined to create diversity in the antibody sequences?
 - (a) Only two DNA regions, V and D, combine in heavy chains
 - (b) V, D, and JH regions are combined in light chains
 - (c) V and JH regions are combined, while D regions are excluded from heavy chains
 - (d) Five separate DNA regions are combined in heavy chains, adding extra complexity
 - (e) D and JH regions are combined, while V regions are excluded from heavy chains
4. What is the probability of getting exactly three heads in five flips of a balanced coin?
 - (a) 12.5 percent
 - (b) 40 percent
 - (c) 18.75 percent
 - (d) 25 percent
 - (e) 37.5 percent
5. If X has a normal distribution with mean 9 and standard deviation 3, find $P(5 < X < 11)$.
 - (a) - .09176

- (b) .21185
- (c) .37748
- (d) .55962
- (e) .65361

True / False (10 marks)

Answer True or False

6. True or False: Directional selection can lead to distinct color patterns in species, as seen in certain African butterflies.
7. True or False: Natural selection does not involve the inheritance of both 'fit' and 'unfit' genes within a population.
8. True or False: The lamb's following behavior can be attributed to social learning, as it observes and imitates Mary's actions.
9. True or False: Clathrin is not involved in the motility of cultured fibroblasts, which instead utilize adhesion plaques, vinculin, and lamellipodia.
10. True or False: Acetylcholine will not be degraded in the synaptic cleft due to the denaturation of enzymes responsible for its breakdown.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how variations in the size of the human appendix among individuals challenge the notion that it is a completely vestigial organ.
12. Discuss the concept of a flock of birds as an elementary form of society, known as a motion group, and analyze the implications of this classification in the study of animal behavior.

End of Paper